Hamza Tahboub

B.S. Computer Science & Mathematics

Boston, MA
tahboub.h@northeastern.edu
www.hamzatahboub.com
hamzatahboub
htahboub

Education

2021 – 2025 (Aug. – Dec.) **B.S. in Computer Science and Mathematics**, *Northeastern University*, Boston

GPA: **3.96/4.0**, Summa Cum Laude

Relevant Coursework: Theory of Computation, Practical Neural Networks, Advanced Linear Algebra, Statistics and Stochastic Processes, Group Theory, Matrix Methods in Machine Learning

Publications

[1] **Hamza Tahboub**, Weiyan Shi, Gang Hua, and Huaizu Jiang. SocialFusion: Addressing Social Degradation in Pre-trained Vision-Language Models. *Transactions on Machine Learning Research*, 2026. arXiv:2512.01148.

Experience

Research

2026 – (Jan.) **Socially coherent multi-person video generation**, *Advised by Professor Huaizu Jiang*, Visual Intelligence Lab, NEU

- Leading a plan-then-render approach to multi-person video generation, targeting the failure of current text-to-video models (including human-centric ones) to produce socially coherent interactions.
- Built a VLM-based social planner that jointly generates each person's motion, emotion, and gaze over the full sequence, extending SocialGen-style motion–language modeling.
- Designed a learned critic to select the best of several candidate plans before conditioning a DiT renderer, scaling compute in plan space rather than pixel space.
- Outperforms baseline video generators on human and social modeling on SVBench, EmoVid, and in a user study at comparable overall video quality. To be submitted to CVPR 2027.

2025 (Feb.) – 2025 (Dec.) **SocialFusion: Unifying visual social interaction understanding**, *Advised by Professors Huaizu Jiang, Weiyan Shi, and Gang Hua*, Visual Intelligence Lab, NEU

- Led the project to unify diverse visual social interaction understanding tasks under one model that leverages synergies between tasks.
- Revealed and diagnosed "social degradation": via representation probing and gradient-conflict analysis, showed that VLM pre-training compresses visual features in ways that erode atomic social cues, causing negative transfer when tasks are trained jointly.
- Proposed SocialFusion, which achieves positive transfer across all five tasks and outperforms three leading VLMs of similar size on four of five. Published in TMLR.

2024 (Jan.) – 2025 (Jan.) **Egocentric Werewolf strategy classification and utterance prediction**, *Advised by Professors Huaizu Jiang and Weiyan Shi*, Visual Intelligence Lab, NEU

- Led this project, which eventually evolved into the above SocialFusion project, to understand subtle social cues from an egocentric perspective.
- Improved strategy prediction over prior methods by exploiting the camera-wearer's attention cues available only from the egocentric view.

2023 Jul. – 2023 Dec. **Genentech R&D Co-op (NLP and Computer Vision)**, *Advised by Dr. Michael Wu and Evan Liu*, San Francisco

- Fine-tuned ensembles of language models and named-entity recognition/relation extraction models on large-scale in-house medical data.
- The resulting system was deployed company-wide for scientists' medical QA and retrieval.
- Implemented state-of-the-art nuclei semantic segmentation methods for in-house use cases and contributed to new approaches.

2022 Aug. – 2023 Aug. **Visual common sense understanding**, *Advised by Professors Byron Wallace and Huaizu Jiang*, Visual Intelligence Lab, NEU

- Found that chain-of-thought prompting hurt smaller language models on visual commonsense QA, contrary to the prevailing view at the time; documented in a manuscript (link).

Teaching

2025 Aug. – 2025 Dec. **Teaching Assistant for Program Design and Implementation**, *Northeastern University*, Boston

- TA for the second course of Northeastern's newly redesigned intro sequence; conducted labs, held office hours, graded, and helped adapt and refine course materials.

2022 Jan. – 2022 Jul. **Teaching Assistant for Fundamentals of Computer Science I**, *Northeastern University*, Boston

- Mentored students for two semesters as a TA for this course taught in Racket (lisp dialect), helping them in office hours, grading their work, and co-leading labs.

Honors and Awards

2024 Aug. – 2024 Dec. **Full-Time Research Fellowship**, semester-long fully-funded research position at the Visual Intelligence Lab.

2024 Jan. – 2024 May. **PEAK Summit Research Award**, pinnacle award for semester-long research in the Northeastern PEAK Fellowship program, including $3,000 of funding for my research.

2021 Aug. – 2025 Dec. **Dean's List**, awarded all semesters at Northeastern University.

Skills

Languages Python, Java, C++, SQL, LaTeX, Lisp, MATLAB, C#

Tools PyTorch, TensorFlow, Git, WandB, NumPy, Pandas, Matplotlib, Scikit-learn